Lecture 5

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Lecture 5

1. More About Distributions and the Normal Curve
2. Markov's Inequality
3. Chebychev's Inequality
4. Reading Summary – Chapter 8
5. Reading Summary – Chapter 9

Finish reading Chapters 8 and 9 for next lecture. Begin reading Chapters 10-12.

Quiz 2 is tomorrow in lab, and Quiz 3 next Wednesday in lecture.

Mean, SD, Percentiles, and Histograms (Normal Curve)

- If the data follow the normal curve. . .
  - The mean and SD tells you everything you need to know.
  - With the mean and SD, you can draw the histogram.
  - With the mean and SD, you can figure out all the percentiles.

Example 1

- Suppose that the histogram of weights for house cats follows the normal curve.
- The mean weight is 9 pounds and the standard deviation 3 pounds.
- Approximate the percentage of cats who are:
  - between 6 and 12 pounds,
  - between 3 and 15 pounds,
  - between 8 and 11 pounds, and
  - exactly 10 pounds.

Example 2

- Find the following percentiles for the weight of house cats (mean: 9 pounds, SD: 3 pounds.)
  - 25th
  - 50th
  - 75th
  - 99th

Mean, SD, Percentiles, and Histograms (Not Normal Curve)

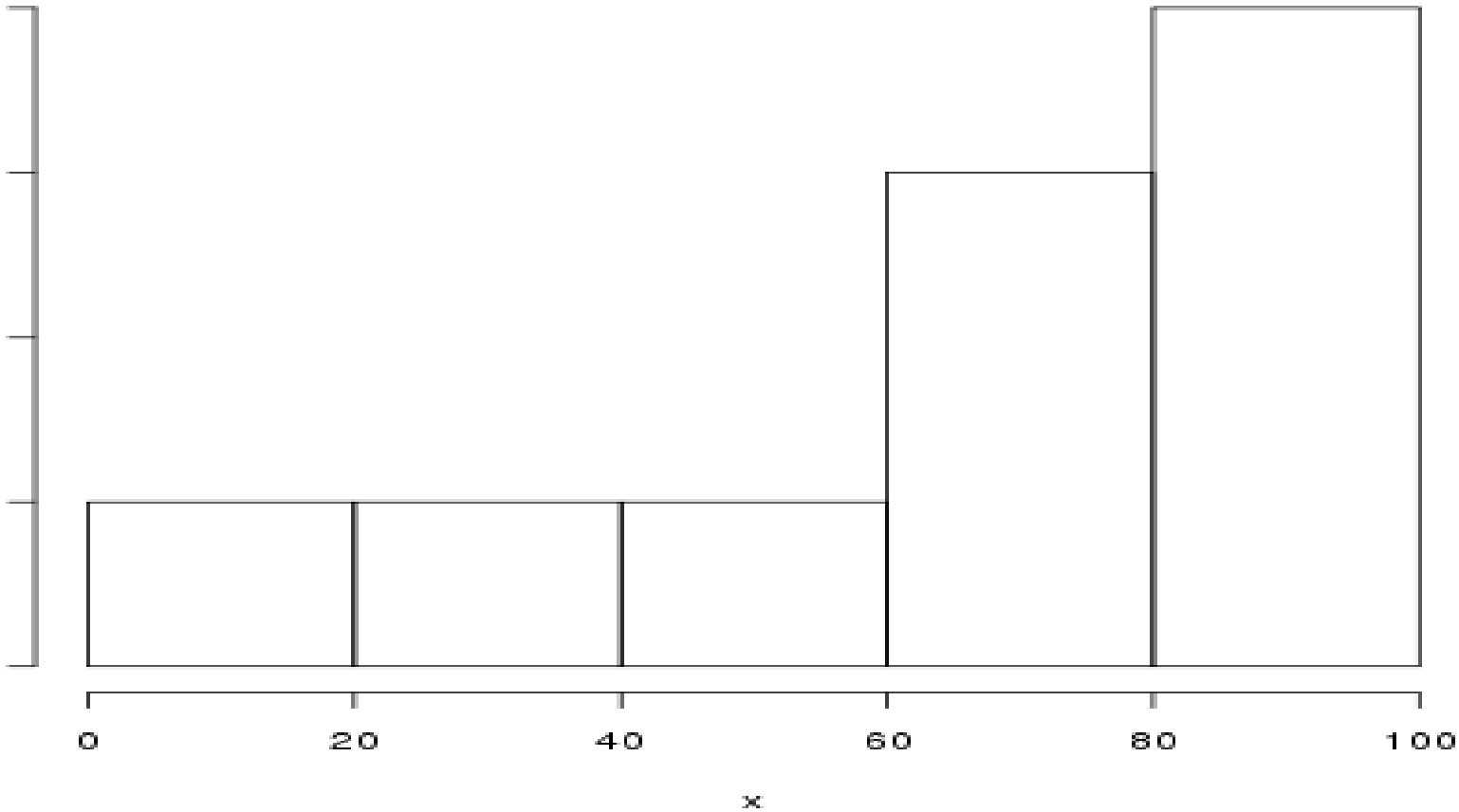
- Whether or not the data follow the normal curve. . .
  - You can draw the histogram if you know the percentiles (approximately).
  - You can figure out the percentiles if you know the histogram (approximately).
  - Knowing histogram $\longleftrightarrow$ knowing percentiles
- If the data don't follow the normal curve. . .
  - The relationship between the mean, SD, percentiles, and the histogram is more complicated.

Relating the Mean and Median to the histogram

- If the histogram is symmetric, the mean equals the median
- If the histogram has a long right tail, the mean is greater than the median
- If the histogram has a long left tail, the mean is less than the median
- Remember:
  - The mean is the balancing point of the histogram.
  - The median is where half the area is to the right, and half is to the left

Discussion Question

- In the histogram below, which is greater: mean or median?
- What is the median? Assume the data is spread evenly within blocks.
- Is the mean greater than or less than 50?



Relating the SD to the IQR

- The IQR can be bigger or smaller than the SD.
- Typically the IQR is bigger than the SD.
  - If the data follow the standard normal curve (mean 0, SD 1), then what is the IQR?
- The SD can be bigger if there are outliers.

Markov's Inequality

Warm-up Discussion Question

- Suppose a list contains only non-negative numbers. The 90th percentile of the list is 25. What is the smallest possible value for the mean?

Answer

- The smallest possible value for the mean is 2.5.
  - This is when 10% of the numbers are 25, and the other 90% of the numbers are 0.

Markov's Inequality

- Markov's inequality takes the same idea as in the previous discussion question, and turns it around.
- The question: Suppose a list of numbers contains only non-negative numbers, and the mean of the list is m.
  - What is the largest possible value of the q-th percentile?
- Answer:

$$\frac{m}{1 - \frac{q}{100\%}}$$

Markov's Inequality – Standard Phrasing

- Suppose a list of numbers contains only non-negative numbers, and the mean of the list is m.
- If x is any number, then:

$$(\text{fraction of list greater than or equal to } x) \leq \frac{m}{x}$$

Discussion Question

- Suppose the average yearly income is $\$60,000$. What is the largest possible fraction of yearly incomes that are greater than $\$120,000$?
- Is your answer realistic? If not, is it too big or too small? Why?

Chebychev's Inequality

- Suppose a list of numbers has mean m, and x is any positive number. Then:

$$(\text{fraction } x \text{ standard deviations or more from } m) \leq \frac{1}{x^2}$$

- Example: The average score on a midterm is 60, and the SD is 15. What is the largest possible fraction of scores that are either greater than 90 or less than 30?
  - Answer: 1/4. Reason: 90 is 2 SDs more than the mean, and 30 is 2 SDs less than the mean.
  - Thus the largest possible fraction of scores greater than 90 or less than 30 is $\frac{1}{2^2}$.

Normal Curve Chebychev Comparison

- Both the normal approximation and Chebychev's inequality address the question:
  - What fraction of the list is more than x standard deviations from the mean?
- The normal approximation gives an approximate answer, but this approximation is only good when the data follow the normal curve.
- Chebychev's inequality doesn't give an approximate answer, but rather a **bound.**
  - The bound is valid for any list, not just lists that follow the normal curve.

Normal Curve vs Chebychev

Number of standard deviations	Normal Curve	Chebychev's Inequality
1	32%	100%
2	5%	25%
3	0.3%	11%
4	0.006%	6%
5	0.00006%	4%
6	0.0000002%	3%

Reading Summary – Chapter 8

Section 8.1: The Scatter Diagram

- Introduction to scatter plots
- **Association**: There is an association between two variables if knowing one variable helps you predict the other.

Section 8.2: The Correlation Coefficient

- **Point of averages**: Center of cloud
- **Correlation Coefficient**: Measure of linear association (clustering around a line)
- Correlation is a number between -1 and 1
- Figure 6: Shows different correlations

Section 8.3: The SD Line.

- The SD line is the line around which the points visually cluster.
- When x increases by 1 SD, y increases by 1 SD. The slope is $SD(y)/SD(x)$.
- Not commonly used.

Section 8.4: Computing the Correlation Coefficient

- Convert the values in each list to standard units. Then compute the products and take the average.
- To see why correlation measures linear association, think about quadrants of the graph.
- Points in the first and third quadrant contribute positively to the correlation.
- Points in the second and fourth contribute negatively.

Reading Summary – Chapter 9

Section 9.1: Features of the Correlation Coefficient

- Correlation is not affected by changes in scale
- Adding a constant to one of the variables will not change the correlation coefficient.
- Multiplying one of the variables by a positive constant will not change the correlation coefficient.

Section 9.2: Changing SDs

- To estimate the correlation visually, you must mentally re-scale the plot so the standard deviations of x and y match up.

Section 9.3: Some Exceptional Cases

- The correlation coefficient is mainly useful for football-shaped data.
- Outliers can make correlation misleading.
- Correlation only measures linear association, not association in general.

Section 4: Ecological Correlations

- **Ecological correlations** are correlations based on rates or averages.
- Ecological correlations can overstate the strength of an association.

Section 5: Association is not Causation

- Correlation only measures association, and association is not causation.
- Associations can be the result of confounding.

Review Questions

- See Review_questions5-6.pdf for review questions for this lecture and next.